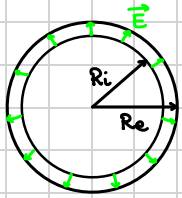
Esercizi scheda 4

Esercizio 1

$R_i = 2{,}2\ mm = 2{,}2 \cdot 10^{-3} m$ $\qquad R_e = 3{,}5\ mm = 3{,}5\ 10^{-3} m$

$l = 2{,}8\ m$ $\qquad \Delta V = 350\ mV = 0{,}35\ V$



$$\Delta V = \frac{\lambda}{2\pi\varepsilon_0} \ln(x_e/x_i) \qquad C = \frac{Q}{\Delta V} = \frac{Q}{\frac{\lambda}{2\pi\varepsilon_0}\ln(x_e/x_i)} = \frac{\cancel{Q}}{\frac{\cancel{Q}}{2\pi\varepsilon_0 L}\cdot \ln(x_e/x_i)} = \frac{2\pi\varepsilon_0 L}{\ln(x_e/x_i)}$$

$$\Rightarrow \frac{C}{L} = \frac{2\pi\varepsilon_0}{\ln(x_e/x_i)} = 1{,}198 \cdot 10^{-10}\ F/m$$

$$C = \frac{C}{L}\cdot L = 3{,}353 \cdot 10^{-10}\ F$$

$Q = C\Delta V$ $\qquad$ La superficie di un cilindro è pari a: $S = 2\pi x L \Rightarrow \sigma_i = \frac{Q}{S_i} = \frac{C\Delta V}{2\pi x_i L} = 3{,}032\ C/m^2$

$$\Rightarrow \sigma_e = \frac{Q}{S_e} = \frac{C\Delta V}{2\pi x_e L} = 1{,}906\ C/m^2$$

Esercizio 2

$\sigma = \pm 0{,}5 \cdot 10^{-3}\ C/m^2$ $\qquad d = 5\ mm = 5 \cdot 10^{-9} m$ $\qquad \varepsilon_r = 5{,}4$ $\qquad \bar{V} = 10^{-16} m^3$

$$E = \frac{\sigma}{\varepsilon} = \frac{\sigma}{\varepsilon_0 \varepsilon_r} = 10{,}46 \cdot 10^6\ N/C$$

La membrana esterna caricata positivamente ha potenziale maggiore. $\Delta V = Ed = 0{,}052\ V$

$$V_{guscio} = \frac{4}{3}\pi x_e^3 - \frac{4}{3}\pi x_i^3 = \frac{4}{3}\pi(x_e^3 - x_i^3) = \frac{4}{3}\pi(x_e - x_i)(x_e^2 + x_i^2 + x_i x_e)$$

$$= \frac{4}{3}\pi \underbrace{(x_e - x_i)}_{=d} \cdot 3 x_e^2 = 4\pi d x_e^2$$

$$= 4\pi d \left(\frac{3\bar{V}}{4\pi}\right)^{\frac{2}{3}}$$

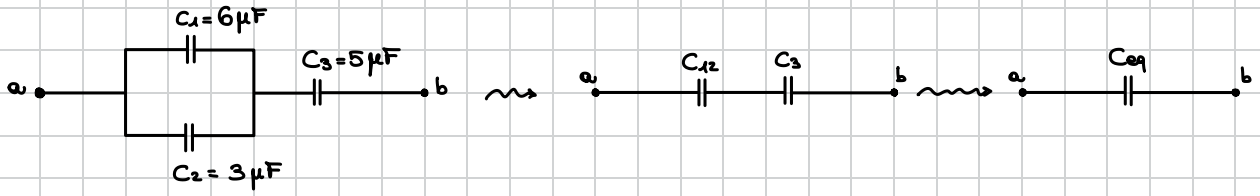
$x_i \sim x_e$

$$\bar{V} = \frac{4}{3}\pi x_e^3 \Rightarrow x_e = \left(\frac{3\bar{V}}{4\pi}\right)^{\frac{1}{3}}$$

$$U = u\, V_{guscio} = \frac{1}{2}\varepsilon E^2 V_{guscio} = \frac{1}{2}\varepsilon_r \varepsilon_0 \frac{\sigma^2}{(\varepsilon_r \varepsilon_0)^2} \cdot 4\pi d \left(\frac{3\bar{V}}{4\pi}\right)^{\frac{2}{3}} = \frac{2\pi d \sigma^2}{\varepsilon_r \varepsilon_0}\left(\frac{3\bar{V}}{4\pi}\right)^{\frac{2}{3}} = 6{,}32\ 10^{-15} J$$

Esercizio 3

Costruisco lo schema equivalente



$C_{12} = C_1 + C_2 = 9\,\mu F$

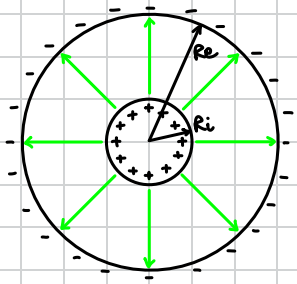
C_1 e C_2 sono in parallelo $\Rightarrow \Delta V_1 = \Delta V_2 \Leftrightarrow \frac{Q_1}{C_1} = \frac{Q_2}{C_2} \Leftrightarrow Q_1 = Q_2 \frac{C_1}{C_2} = 60\,\mu C$

$Q_3 = Q_1 + Q_2 = 90\,\mu C = Q_{eq}$

$$\frac{1}{C_{eq}} = \frac{1}{C_{12}} + \frac{1}{C_3} \Rightarrow C_{eq} = \frac{C_{12} C_3}{C_{12} + C_3} \Rightarrow Q_{eq} = V_{ab} \cdot C_{eq} \Rightarrow V_{ab} = \frac{Q_{eq}(C_{12}+C_3)}{C_{12} C_3} = 28V$$

Esercizio 4

Per praticità, assumiamo una coordinata radiale attraverso cui calcolare la differenza di potenziale.



$\Delta V = V_e - V_i$

$$\Rightarrow \Delta V = -\int_{R_i}^{R_e} E(r)\,dr = -\int_{R_i}^{R_e} \frac{Q}{4\pi\varepsilon_0} \cdot \frac{1}{r^2} = \frac{Q}{4\pi\varepsilon_0} \int_{R_i}^{R_e} -\frac{1}{r^2}\,dr = \frac{Q}{4\pi\varepsilon_0} \left[\frac{1}{r}\right]_{R_i}^{R_e} =$$

$$= \frac{Q}{4\pi\varepsilon_0}\left(\frac{1}{R_e} - \frac{1}{R_i}\right) = \frac{Q(R_i - R_e)}{4\pi\varepsilon_0 \cdot R_i R_e}$$

$$= -\frac{Q(R_e - R_i)}{4\pi\varepsilon_0 R_i R_e}$$

$$C = \frac{Q}{\Delta V} = \frac{\cancel{Q}}{-\frac{\cancel{Q}(R_e - R_i)}{4\pi\varepsilon_0 R_i R_e}} = -\frac{4\pi\varepsilon_0 R_i R_e}{R_e - R_i}$$

Analisi dimensionale del membro di sinistra:

so che $\Delta U = q\Delta V \Rightarrow 1V = \frac{1J}{1C}$ $\qquad \frac{J}{C} = kg\frac{m^2}{s^2} \cdot \frac{1}{C} = \frac{kg\,m^2}{C s^2} \Rightarrow \left[\frac{Q}{\Delta V}\right] = \frac{C}{\frac{kg\,m^2}{C s^2}} = \frac{C^2 s^2}{kg\,m^2}$

Analisi dimensionale del membro di destra:

$$\left[-\frac{4\pi\varepsilon_0 R_i R_e}{R_e - R_i}\right] = \frac{1 \cdot 1 \cdot \left(N\frac{m^2}{C^2}\right)^{-1} \cdot m \cdot \cancel{m}}{\cancel{m}} = \frac{C^2}{Nm^2} \cdot \cancel{m} = \frac{C^2}{kg\frac{m}{s^2} \cdot m} = \frac{C^2 s^2}{kg\,m^2}$$

VERIFICATO